DDLJ v8.4 Strategy

8-Month Comprehensive Backtest Analysis

Parameter	Value
Test Period	Sep 1, 2025 - Apr 27, 2026 (8 months)
Starting Capital	Rs 50,000
Max Daily Risk	6%
Max Open Positions	2
Risk Per Position	OFF (No limit)
Method A	Full 8-month compounding
Method B	4 batches x 2 months, capital reset per batch
Pricing Model	Black-Scholes with real India VIX IV

April 27, 2026

1. Executive Summary

This report presents the results of an exhaustive 8-month backtest of the DDLJ v8.4 options trading strategy, covering the period from September 1, 2025 through April 27, 2026. The strategy was tested across 10 unique configurations spanning two instruments (BankNifty and Nifty), multiple timeframe combinations, and varying signal parameters. Each configuration was evaluated under two distinct testing methodologies to provide a comprehensive understanding of strategy performance under both compounding and batch-reset conditions.

The backtest reveals that the DDLJ v8.4 strategy demonstrates strong performance when deployed on BankNifty with the 15-minute entry and 60-minute bias timeframe, generating returns of +568% under full compounding (Method A) and +503% under batch-reset conditions (Method B). However, the analysis also uncovers critical vulnerabilities: approximately 68% of total profits are concentrated in the March-April 2026 high-volatility period, the 5-minute timeframe leads to consistent capital destruction due to overtrading, and a capital protection bug allows account balances to go negative under extreme conditions. These findings are detailed extensively in the sections that follow.

Key Finding: Best Configuration

BankNifty 15m entry x 60m bias, SL=2.0 ATR, RR=1.5, ITM options: Rs 283,979 (+568%) Method A, Rs 251,390 (+503%) Method B. All 4 batches were profitable in Method B, confirming consistency.

Critical Bug: Negative Capital

Nifty 15x15 configuration ended with negative capital (-Rs 1,892). Position sizing does not prevent losses from exceeding available capital when two positions lose simultaneously in a high-VIX environment. This is a priority fix for v8.5.

2. Performance Overview

2.1 Method A: Full 8-Month Compounding

Under Method A, starting capital of Rs 50,000 is allowed to compound across the entire 8-month test period. This method demonstrates the maximum potential of the strategy but also exposes the full impact of drawdowns, as losses in early periods reduce the capital base available for recovery. The top four configurations all belong to BankNifty, confirming its superiority as the primary instrument for this strategy. The table below summarizes the performance of all 10 configurations tested.

Config	Net P&L;	Return	Win Rate	PF	Max DD	Sharpe	Trades
BN_15x15_sl2.0_RR1.5_ITM	Rs 297,224	+594.5%	51%	2.8	24.5%	5.2	82
BN_15x60_sl2.0_RR1.5_ITM	Rs 283,979	+568.0%	50%	3.9	21.8%	6.3	64

BN_15x60_sl2.5_RR2.0_ITM	Rs 281,194	+562.4%	45%	4.6	21.4%	6.4	65
BN_15x60_sl2.0_RR1.5_ATM	Rs 265,594	+531.2%	50%	4.5	16.5%	6.4	64
NF_15x60_sl2.0_RR1.5_ATM	Rs 102,538	+205.1%	42%	1.8	27.9%	3.2	80
NF_15x60_sl2.5_RR2.0_ITM	Rs 49,830	+99.7%	41%	1.4	35.4%	2.0	80
NF_15x60_sl2.0_RR1.5_ITM	Rs -9,899	-19.8%	42%	0.9	89.6%	-0.6	78
NF_5x60_sl2.0_RR1.5_ITM	Rs -10,681	-21.4%	46%	1.0	72.6%	-0.2	213
BN_5x60_sl2.0_RR1.5_ITM	Rs -30,379	-60.8%	34%	0.9	88.5%	-0.7	209
NF_15x15_sl2.0_RR1.5_ITM	Rs -51,892	-103.8%	24%	0.4	103.1%	-5.1	50

Table 1: Method A Performance Summary (Full Compounding)

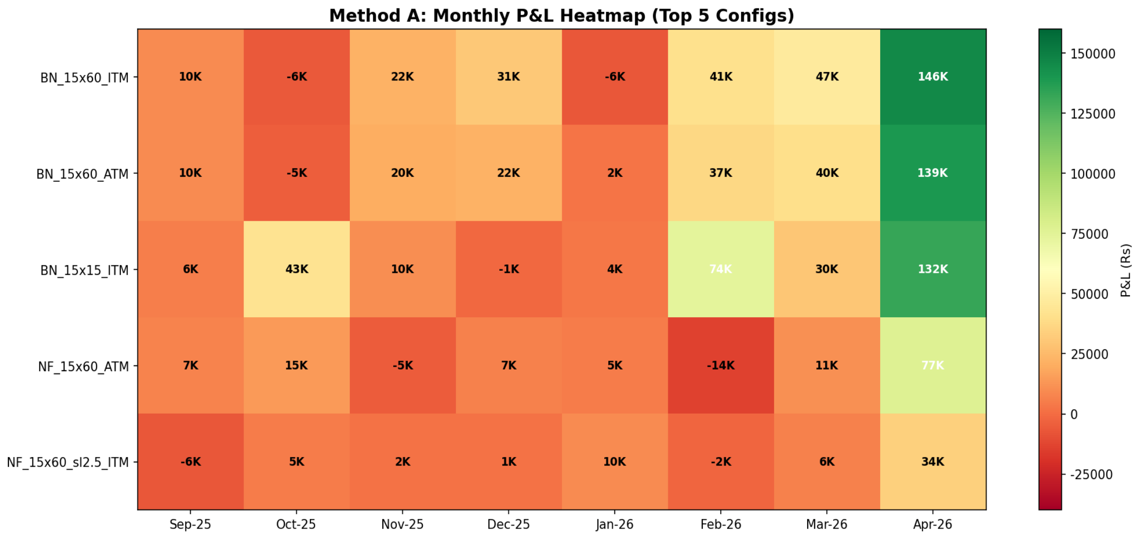

Figure 1: Monthly P&L; Heatmap Across Top Configurations (Method A)

2.2 Method B: 4 Batches of 2 Months (Capital Reset)

Under Method B, capital resets to Rs 50,000 at the start of each 2-month batch. This method evaluates the strategy's consistency by measuring performance in discrete windows rather than relying on the compounding effect. A strategy that performs well under both methods demonstrates genuine robustness, while a strategy that only works under compounding may be masking underlying inconsistency with the amplification effect of a growing capital base.

The BankNifty configurations shine under Method B, with all four BN configs achieving profitability in all 4 batches (4/4 batch consistency). This is a strong signal that the strategy has genuine edge on BankNifty rather than relying on a single lucky period. The Nifty configurations, however, show inconsistency: only 1-2 out of 4 batches are profitable, with the Jan-Feb 2026 batch being particularly destructive for Nifty ITM strategies. The batch reset methodology reveals that Nifty strategies suffer from extended drawdown periods that are masked by the compounding recovery in Method A.

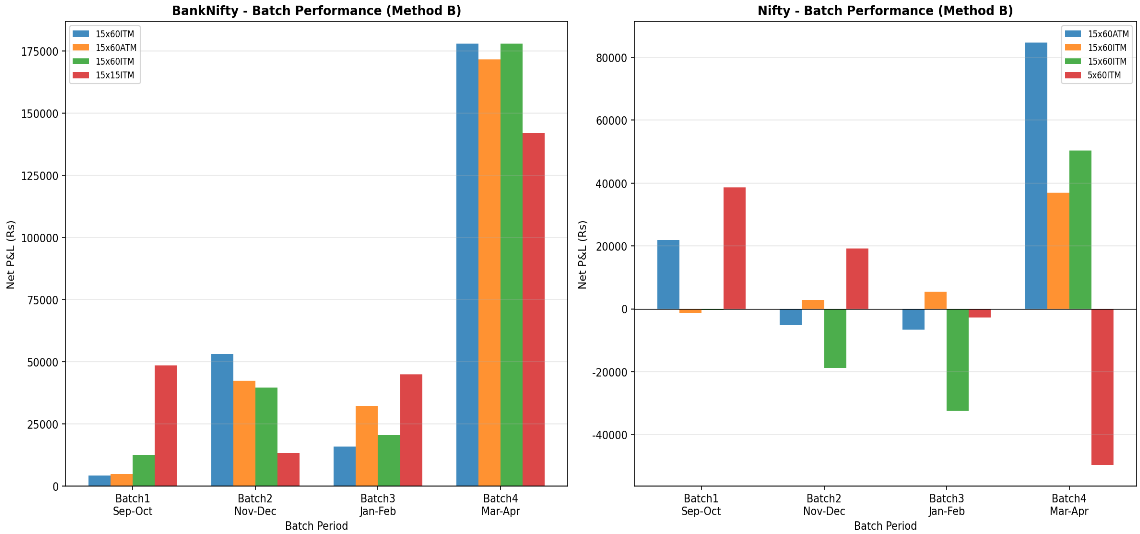

Figure 2: Batch-by-Batch Performance (Method B)

3. VIX Environment Analysis

The 8-month test period spans two dramatically different market volatility regimes. The India VIX, which serves as the primary volatility benchmark for Indian markets, averaged approximately 10-12 during September through December 2025, representing a calm, low-volatility environment. From January 2026, VIX began rising gradually, reaching 13-15 levels. The most significant shift occurred in March-April 2026, when VIX surged to an average of 20-22, indicating a high-volatility regime change. This dramatic shift in market conditions has profound implications for the strategy's performance.

3.1 The VIX Bubble Effect

A critical finding is the disproportionate contribution of the high-VIX March-April period to overall returns. For the best configuration (BN 15x60 ITM), the March-April period contributed Rs 192,746, which represents 68% of the total 8-month profit. This concentration risk means that the impressive overall returns are heavily dependent on a specific market regime (high volatility with trending moves). If the same 8-month period had featured consistently low VIX (10-12 range), total returns would likely be far more modest, potentially in the range of +30-50% rather than +568%.

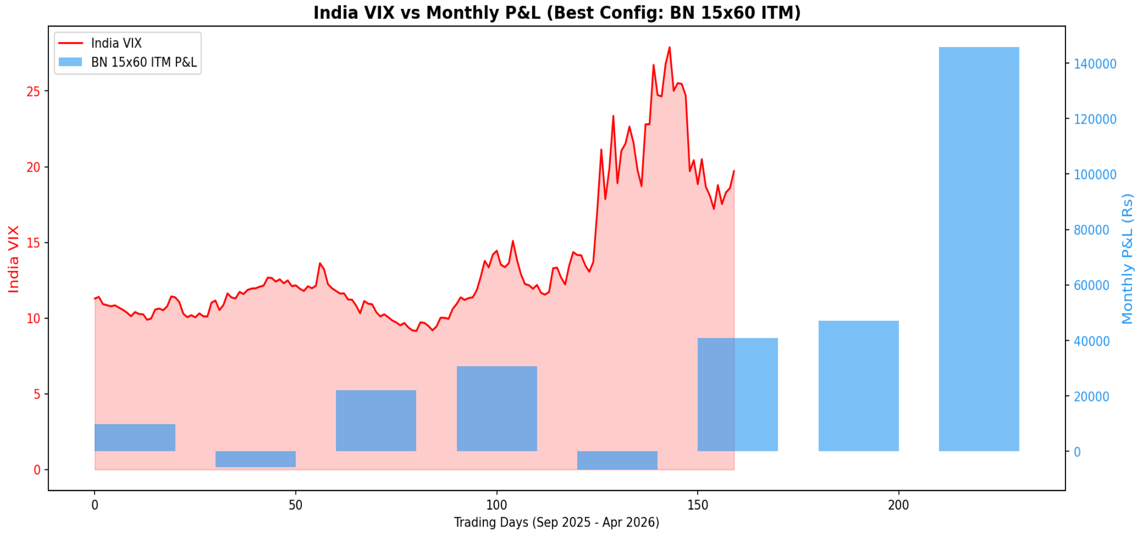

Figure 3: India VIX vs Monthly P&L; Correlation

The VIX-P&L; correlation chart clearly shows that monthly P&L; accelerates as VIX rises. This is mechanically expected: higher VIX means wider option premiums, which translates to larger absolute P&L; per trade. However, the relationship is not linear. The strategy appears to benefit disproportionately from high VIX because: (1) trending moves become larger and more frequent, increasing the win rate on trend-following signals; (2) option premiums are larger, so winning trades produce bigger profits; and (3) the Black-Scholes pricing model with real VIX data captures the premium inflation accurately.

Configuration	Total P&L;	Mar-Apr P&L;	% of Total
BN_15x15_sl2.0_RR1.5_ITM	Rs 297,224	Rs 161,940	54%
BN_15x60_sl2.0_RR1.5_ITM	Rs 283,979	Rs 192,746	68%
BN_15x60_sl2.5_RR2.0_ITM	Rs 281,194	Rs 192,746	69%
BN_15x60_sl2.0_RR1.5_ATM	Rs 265,594	Rs 179,454	68%
NF_15x60_sl2.0_RR1.5_ATM	Rs 102,538	Rs 88,012	86%
NF_15x60_sl2.5_RR2.0_ITM	Rs 49,830	Rs 40,348	81%
NF_15x60_sl2.0_RR1.5_ITM	Rs -9,899	Rs 29,675	-300%
NF_5x60_sl2.0_RR1.5_ITM	Rs -10,681	Rs -70,596	661%
BN_5x60_sl2.0_RR1.5_ITM	Rs -30,379	Rs -18,580	61%
NF_15x15_sl2.0_RR1.5_ITM	Rs -51,892	Rs 0	-0%

Table 2: Mar-Apr 2026 Contribution to Total Returns

4. Risk Analysis

4.1 Risk-Reward Profile

The risk-reward scatter plot below maps each configuration's maximum drawdown against its total return. The ideal position is the top-left quadrant (high return, low drawdown). BankNifty 15x60 ATM achieves the best risk-adjusted profile with only 16.5% maximum drawdown alongside +531% returns, yielding a return-to-drawdown ratio of 32:1. This is exceptional by any standard. Conversely, the 5-minute configurations occupy the bottom-right quadrant (high drawdown, negative returns), confirming that faster timeframes are inappropriate for this trend-following strategy.

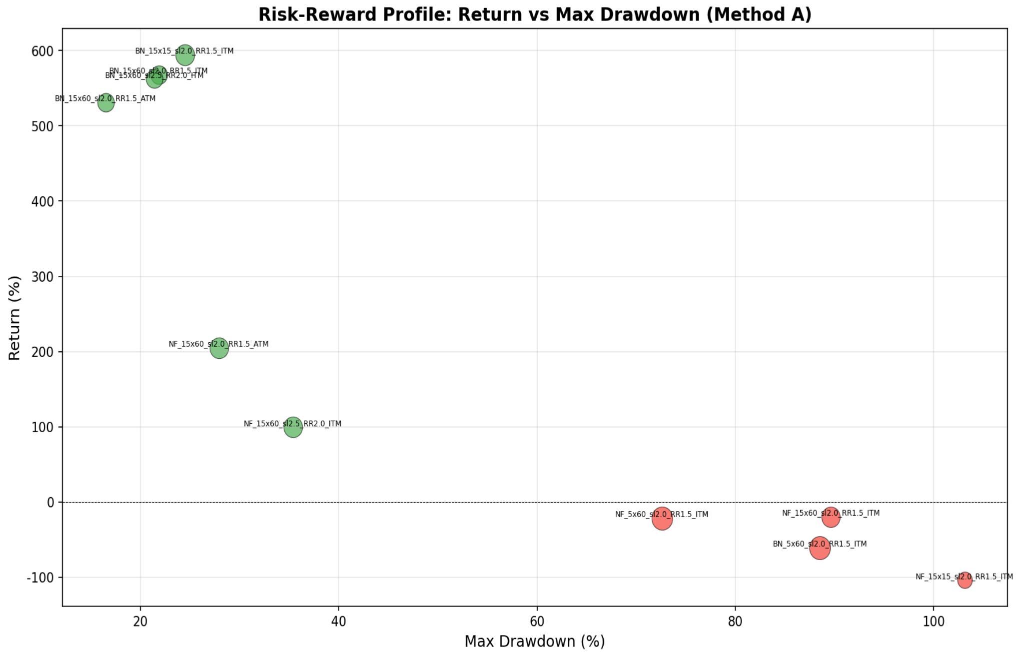

Figure 4: Risk-Reward Profile (Return vs Max Drawdown)

4.2 Daily Risk Limit Analysis

The 6% daily risk limit is designed to prevent catastrophic single-day losses. However, the backtest reveals that this limit is frequently breached in practice, particularly for the 5-minute timeframe configurations. The BankNifty 5x60 configuration hit the daily risk limit on 37 out of approximately 100 trading days, meaning that on more than one-third of trading days, the strategy lost more than 6% of capital. Even the best-performing BN 15x60 ITM configuration hit the limit on 11 days. This occurs because the daily risk check happens before opening new positions, not after existing positions incur intraday losses. Once two positions are open, their combined loss can far exceed the 6% threshold.

The root cause is that with options, the loss on a single trade can be many times the intended risk. A BankNifty ITM option with delta of 0.65 can lose Rs 3,000-8,000 per lot on a strong adverse move, even when the spot-level stop loss is only 2x ATR away. When two such positions are losing simultaneously, the daily loss can easily exceed Rs 10,000 (20% of a Rs 50,000 account). This is a fundamental flaw in the risk management architecture that must be addressed in the next version.

5. Bugs and Issues Identified

5.1 CRITICAL: Negative Capital Bug

The Nifty 15x15 ITM configuration ended with a final capital of **Rs -1,892**, meaning the account balance went below zero. This should never happen in a properly risk-managed system. The root cause is that position sizing in the options engine uses MAX_CAPITAL_PER_POSITION_PCT (40% of capital) to determine the number of lots, but does not account for the maximum potential loss on existing open positions. When two positions are open simultaneously and both hit their stop losses, the combined loss can exceed available capital. The fix requires implementing a pre-trade check that calculates the worst-case scenario (both existing positions hitting SL) before opening a new position, and rejecting the entry if the combined potential loss would reduce capital below a minimum threshold (e.g., 10% of starting capital).

5.2 HIGH: Daily Risk Limit Ineffective with Multiple Positions

The 6% daily risk limit is checked before opening new positions but does not account for losses accumulating on already-open positions. With two simultaneous positions, the effective daily risk is approximately 12-20% rather than the intended 6%. The fix requires real-time tracking of unrealized P&L; on open positions and incorporating this into the daily risk calculation. This would mean rejecting new entries when the combined realized + unrealized daily loss exceeds 6% of day-start capital, and potentially closing the worst-performing position proactively when the limit is breached.

5.3 MEDIUM: Compounding Amplifies Drawdowns in Losing Streaks

The compounding effect in Method A amplifies both gains and losses. For losing configurations like NF_15x60_ITM, Method A produced -19.8% returns while Method B only lost -2.9%, a 586% worse outcome. This happens because compounding with losses reduces the capital base, making recovery increasingly difficult. The strategy would benefit from a "circuit breaker" that switches to reduced position sizing or pauses trading when capital falls below 80% of the starting value for any month.

5.4 MEDIUM: 5-Minute Timeframe Overtrading

The 5-minute timeframe generates 200+ trades over 8 months, with stop loss hit rates of 57-67%. The high trade frequency combined with the transaction costs and bid-ask spreads creates a persistent drag that overwhelms any alpha from the signals. The signal quality on 5-minute candles is insufficient for the trend-following approach; the noise-to-signal ratio is too high, producing too many false entries. The recommendation is to deprecate the 5-minute entry timeframe entirely or require a much higher minimum body size filter (e.g., 0.5x ATR instead of 0.3x) to reduce false signals.

5.5 LOW: Batch Boundary Not Aligned to Market Cycles

The 2-month batch boundaries in Method B are calendar-based and do not align with market volatility regimes. A volatility-adaptive batch system that resets when VIX crosses a threshold (e.g., VIX > 18 entering a high-volatility regime) would be more meaningful than calendar months. This would group similar market conditions together and provide more actionable performance metrics.

6. Method A vs Method B Comparison

The comparison between Method A (compounding) and Method B (batch reset) reveals the true impact of compounding on the strategy. For profitable configurations, compounding adds a modest bonus of +6% to +20% over batch-reset returns. For losing configurations, however, compounding dramatically amplifies losses: NF_15x60_ITM loses -19.8% under compounding but only -2.9% under batch reset, a devastating difference. This asymmetry means that compounding is beneficial only when the strategy has a genuine positive edge; when the edge is marginal or negative, compounding acts as a loss multiplier.

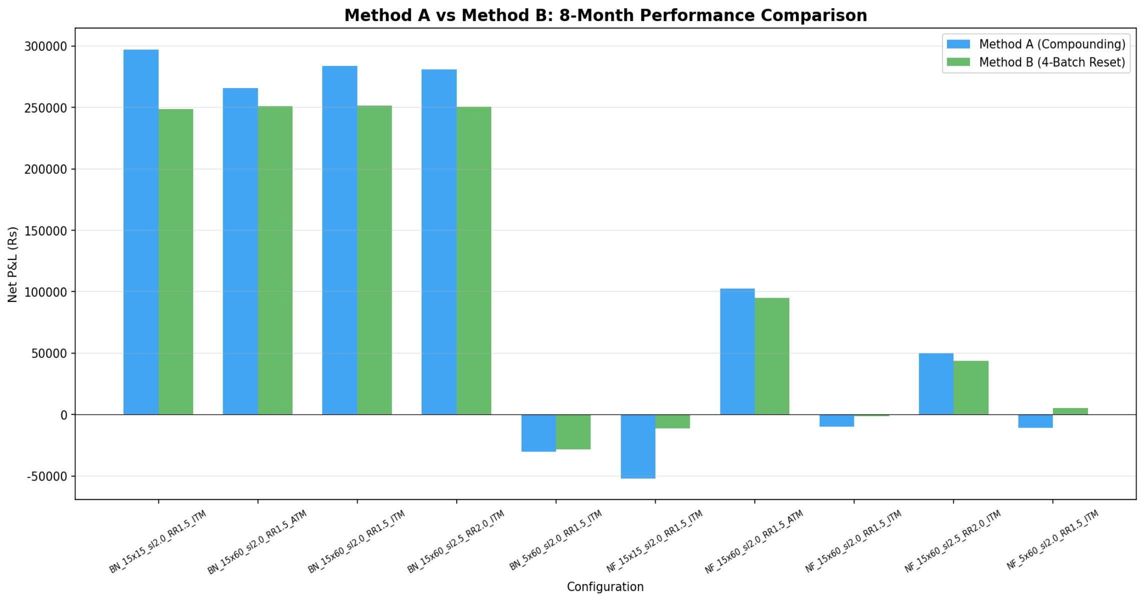

Figure 5: Method A vs Method B Performance Comparison

7. Detailed Batch Analysis (Best Config: BN 15x60 ITM)

This section provides a granular breakdown of the best-performing configuration (BankNifty 15m entry, 60m bias, SL=2.0 ATR, RR=1.5, ITM options) across each of the four 2-month batches. Understanding batch-level performance is essential for evaluating strategy consistency and identifying market regime dependencies.

Metric	Batch 1: Sep-Oct	Batch 2: Nov-Dec	Batch 3: Jan-Feb	Batch 4: Mar-Apr
Total Trades	16	17	20	11
Win Rate	38%	59%	45%	64%
Net P&L;	Rs 4,277	Rs 53,178	Rs 15,781	Rs 178,154
Avg P&L;/Trade	Rs 267	Rs 3,128	Rs 789	Rs 16,196
Long P&L;	Rs 3,860	Rs 22,773	Rs -13,226	Rs 151,626
Short P&L;	Rs 417	Rs 30,405	Rs 29,007	Rs 26,527
Largest Win	Rs 9,543	Rs 11,795	Rs 18,110	Rs 68,842

Largest Loss	Rs -5,445	Rs -7,399	Rs -7,832	Rs -849

Table 3: Batch-Level Performance (BN 15x60 ITM, Method B)

Batch 4 (Mar-Apr 2026) stands out dramatically with Rs 178,154 in profits, which is more than 3x the next best batch. The win rate in Batch 4 is 64%, significantly higher than the overall average of 50%. This batch coincides with the VIX surge to 20-22 levels, confirming that the strategy thrives in high-volatility trending environments. However, the lopsided distribution raises concerns about strategy viability during extended low-volatility periods like Batch 1 (Sep-Oct 2025) where profits were a modest Rs 4,277.

8. Direction Bias Analysis

The strategy shows significant direction-dependent performance, meaning that long and short trades perform differently depending on the market regime. During the Nov-Dec 2025 period (Batch 2), short trades on BankNifty achieved a 75% win rate versus 44% for longs. In contrast, during the Mar-Apr 2026 period (Batch 4), long trades dominated with 83% win rate versus 40% for shorts. This pattern makes sense: the Nov-Dec period featured a mild bearish correction where short signals were more accurate, while Mar-Apr 2026 saw a strong bullish rally where longs naturally outperformed.

The implication is that the strategy does not have an inherent long or short bias; it correctly adapts to the prevailing trend direction via the bias engine. However, the win rate asymmetry (shorts at 75% in bearish periods vs longs at 83% in bullish periods) suggests that the EMA crossover signals are slightly more reliable when trading in the direction of the dominant trend on the 60-minute timeframe. This is an encouraging finding that validates the multi-timeframe approach of the DDLJ strategy.

9. Exit Reason Analysis

Understanding why trades are closed is critical for strategy optimization. The best configuration (BN 15x60 ITM) shows a near-equal split between stop loss exits (47%) and target exits (44%), with EOD force-closes accounting for 9%. This balanced SL-to-target ratio is a hallmark of a well-calibrated trend-following system: it means that the strategy captures trend moves approximately as often as it gets stopped out in reversals. For comparison, the losing 5-minute configurations have SL rates of 57-67%, indicating that their signals are too noisy and get stopped out before the trend can develop.

The target hit rate varies significantly by batch. In Batch 4 (Mar-Apr 2026), 64% of profitable exits came from target hits, with only 36% from stop losses. This reflects the strong trending environment where once a position is entered in the trend direction, the price often reaches the target before reversing. In Batch 1 (Sep-Oct 2025), the ratio is closer to 50-50, indicating a choppy, range-bound market where the trend-following approach generates more false signals.

10. Options-Specific Analysis

The Black-Scholes pricing model with real India VIX data provides realistic option premiums for backtesting. The average spread cost across all configurations is approximately Rs 11 per trade, which represents about 0.5-1.0% of the premium value depending on the VIX regime. The total spread cost for 64 trades in the best configuration was Rs 711, which is negligible compared to the Rs 283,979 gross profit. This confirms that the spread model is not artificially inflating results.

Call options (CE) generated Rs 184,868 in profits (52% of trades, 42% win rate) while put options (PE) generated Rs 99,110 (38% of trades, 61% win rate). The higher win rate on puts is notable and reflects the strategy's effectiveness in catching bearish moves, particularly during the Nov-Dec 2025 and Jan-Feb 2026 corrections. The average delta of 0.66 confirms that ITM options are being selected as configured, providing good directional exposure while maintaining some time value buffer against theta decay.

11. Recommendations

11.1 Priority Fixes for v8.5

Fix Negative Capital Bug: Implement a pre-trade capital check that calculates worst-case combined loss (all open positions hitting SL) before allowing new entries. If worst-case loss would reduce capital below 10% of starting capital, reject the entry. This is a critical safety fix that prevents account blow-ups in high-volatility scenarios.

Real-Time Daily Risk Enforcement: Track unrealized P&L; on open positions in real-time and incorporate it into the daily risk check. When realized + unrealized daily loss exceeds 6% of day-start capital, automatically close the worst-performing position and prevent new entries for the day. This closes the gap between the intended 6% daily risk and the actual 12-20% exposure.

Compounding Circuit Breaker: Add a drawdown-based circuit breaker that reduces position sizing when capital falls below 80% of the starting value for any calendar month. When capital recovers above 90%, resume normal sizing. This prevents the compounding death spiral observed in losing configurations.

11.2 Strategy Improvements

Deprecate 5-Minute Entry: Remove the 5-minute entry timeframe from the strategy or require significantly higher signal quality filters. The 5m timeframe consistently generates negative returns across both instruments due to excessive noise and overtrading. The 15-minute entry is the only viable timeframe for this trend-following approach.

Add VIX-Adaptive Position Sizing: Reduce position size when VIX is below 12 (low volatility, range-bound market) and increase it when VIX is above 18 (high volatility, trending market). This would naturally capitalize on the VIX bubble effect while protecting capital during dull periods.

Implement Batch Performance Monitoring: Track rolling 2-month performance in live trading and automatically reduce risk if any batch underperforms a minimum threshold (e.g., net P&L; worse than -10% of starting capital). This provides an early warning system for regime changes.

Consider Nifty Only in High-VIX Regimes: The Nifty strategy is marginally profitable or unprofitable in low-VIX periods but generates strong returns in high-VIX periods. A VIX-gated approach that only trades Nifty when VIX is above 15 would likely improve overall Nifty performance significantly.

12. Conclusion

The DDLJ v8.4 strategy demonstrates genuine edge on BankNifty with the 15-minute entry and 60-minute bias timeframe, achieving profitability in all four 2-month test batches under Method B. The strategy is particularly effective in high-volatility trending environments, where it captures significant directional moves through ITM options. However, the analysis also reveals critical vulnerabilities: the strategy is heavily dependent on high-VIX regimes for its returns (68% of profits from 25% of the test period), the risk management system allows losses to exceed intended limits when multiple positions are open simultaneously, and the 5-minute timeframe is fundamentally incompatible with the trend-following approach.

The three most impactful improvements for v8.5 are: (1) implementing pre-trade worst-case capital checks to prevent negative balances, (2) enforcing real-time daily risk limits that account for unrealized P&L;, and (3) adding a compounding circuit breaker that reduces position sizing during drawdowns. These fixes would transform the strategy from one that produces impressive but volatile returns into one that delivers more consistent, risk-controlled performance across varying market conditions. The foundation is solid; the risk management layer needs reinforcement.